

MACHINE LEARNING FUNDAMENTALS - LS2026
SEMINAR: PROBABLY APPROXIMATELY CORRET LEARNING

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Assignment 1. Consider a binary classification problem, with the label space $\mathcal{Y} = \{-1, +1\}$, and input space $\mathcal{X} = \mathbb{R}^n$, consisting of n -dimensional real-valued vectors. The goal is to learn a classifier $h: \mathcal{X} \rightarrow \mathcal{Y}$ that minimizes the probability of misclassification. Let $\mathcal{R}_1, \mathcal{R}_2, \dots, \mathcal{R}_M$ be a partitioning of the input space, i.e.

$$\bigcup_{r=1}^M \mathcal{R}_r = \mathcal{X} \quad \text{and} \quad \mathcal{R}_r \cap \mathcal{R}_s = \emptyset \quad \text{for} \quad r \neq s,$$

where each region $\mathcal{R}_r$ corresponds to a leaf of a (fixed) binary decision tree. Define the hypotheses class

$$\mathcal{H} = \left\{ h: \mathcal{X} \rightarrow \mathcal{Y} \mid h(x) = \sum_{r=1}^M c_r \mathbb{I}[x \in \mathcal{R}_r], c \in \{-1, +1\}^M \right\}.$$

Thus, each hypothesis $h \in \mathcal{H}$ is fully specified by a parameter vector $c = (c_1, \dots, c_M) \in \{-1, +1\}^M$, which assigns a class label to each leaf of the tree. The indicator function $\mathbb{I}[x \in \mathcal{R}_r]$ equals 1 if x falls into region $\mathcal{R}_r$, and 0, otherwise. You are given training examples $T_m = ((x_i, y_i) \in \mathcal{X} \times \mathcal{Y} \mid i = 1, \dots, m)$ consisting of m i.i.d. samples drawn from an unknown joint distribution with density $p(x, y)$. Assume the partitioning $\mathcal{R}_1, \mathcal{R}_2, \dots, \mathcal{R}_M$, has been constructed independently of the training examples T_m (e.g., randomly or using a different dataset). The training examples T_m are used solely to assign class labels to the leaves, i.e. to learn the parameter vector c .

- a)** Describe an algorithm based on the Empirical Risk Minimization principle that uses the training examples T_m to learn a classifier h_m from the hypothesis space $\mathcal{H}$. Derive an explicit formula for computing the parameter vector c from the training examples T_m .
- b)** Is the algorithm from a) a successful PAC learning algorithm ? Justify your answer.
- c)** Assume that the class-conditional distributions $p(x \mid y = +1)$ and $p(x \mid y = -1)$ are multivariate normal distributions. In this case, is the approximation error of the algorithm equal to zero? Justify your answer.
- d)** How many training examples m are required to have a guarantee that the algorithm learns a classifier h_m whose estimation error is at most $\varepsilon \in (0, 1)$ with probability at least $1 - \delta \in (0, 1)$? Compute the number of the training examples for $\varepsilon = 0.05$, $\delta = 0.1$ and $M = 128$.
- e)** What is the VC dimension of the hypothesis class $\mathcal{H}$?

Solution 1. a) The classifier that minimizes the empirical error assigns each leaf according to the frequency of labels of training examples that fall to the particular leaf. That is, the

elements of the parameter vector are computed according to:

$$c_r = \text{sign} \left(\sum_{i=1}^m \mathbb{I}[x_i \in \mathcal{R}_r] y_i \right) .$$

b) The hypothesis space $\mathcal{H}$ is finite, containing 2^M hypotheses. As a result, the uniform law of large numbers (ULLN) holds for $\mathcal{H}$. Since the learning algorithm follows the empirical risk minimization (ERM) principle, it is a successful PAC learner.

c) The Bayes classifier for the multivariate normal distributions is a quadratic classifier w.r.t. input $x \in \mathbb{R}^n$. Clearly, not all quadratic classifiers can be modeled by the finite hypothesis class $\mathcal{H}$. As the result, the approximation error is non-zero.

d) For a finite hypothesis space, the sample complexity (i.e. number of training examples required to ensure an estimation error of at most ε with confidence at least $1 - \delta$) is given by:

$$m \geq 2 \frac{\log(2|\mathcal{H}|) - \log(\delta)}{\varepsilon^2} .$$

Since $|\mathcal{H}| = 2^M$, we can rewrite this as

$$m \geq \frac{2}{\varepsilon^2} ((M + 1) \log 2 - \log(\delta)) .$$

For $M = 128$, $\varepsilon = 0.05$, and $\delta = 0.1$, this gives:

$$m \geq \frac{2}{0.05^2} ((128 + 1) \log 2 - \log(0.1)) \approx 73,375 .$$

Hence, approximately 73,375 training examples are needed to learn approximately correct predictor with the desired confidence.

d) The maximal number of inputs that can be shattered equals M . That is, VC dimension of $\mathcal{H}$ is M .

Assignment 2. Assume you want to use a large language model (LLM) to summarize text into a structured output (e.g., a JSON object). To achieve this, you need to design a suitable prompt that effectively instructs the LLM to produce the desired summary format. Suppose you manually (or with the help of another LLM) design a set of 100 candidate prompts. To select the best prompt, you evaluate each one on a benchmark dataset consisting of m examples — each containing an input text and its corresponding ground-truth summary. Assume that you have access to another LLM capable of evaluating whether a generated summary is **good** or **bad** compared to the ground truth.

Knowledge Extraction

PROMPT

Instructions

Task Overview

Your task is to extract each person mentioned in the article and a list of all claims given by that person.

Guidelines

...

Query

<date>čtvrtek 19/12/2024**</date>**
<title>Stenografický zápis 125. schůze, 19. prosince 2024**</title>**
<document>
 Místopředsedkyně PSP Věra Kovářová: Také děkuji. Jenom si tedy potřebuji - děkuju vám, paní zpravodajko - jenom si potřebuji ujasnit vystoupení s přednostním právem pana předsedy Jakuba Michálka...
 Ministr spravedlnosti ČR Pavel Blažek: Děkuji. Já musím zareagovat...
</document>



```
{
  "people": [
    {
      "id": 643,
      "name": "Věra Kovářová",
      "roles": [
        "místopředsedkyně PSP"
      ]
    },
    {
      "id": 174,
      "name": "Pavel Blažek",
      "roles": [
        "ministr spravedlnosti ČR"
      ]
    },
    ...
  ],
  "claims": [
    ...,
    {
      "id": 3,
      "person_id": 643,
      "text": "Podporuji projednání zákona tak, aby mohl nabýt účinnosti 1. ledna 2026.",
      "context": "...",
      "start": 123,
      "end": 195
    },
    ...
  ]
}
```

Prompt example:

To transform a query into an answer, follow these refined steps:

- 1. Identify Main Events:** Extract the primary event(s) from the text, focusing on key actions or occurrences. Each main event should be distinct and significant. For example, in a query about election results and subsequent developments, "Hnutí ANO remains in opposition despite winning the elections" and "Markéta Vaňková becomes the probable mayor" are two distinct main events.
- 2. Extract Subevents:** Identify related subevents or specific details tied to each main event. Subevents should provide additional context or details about the main event. Ensure each subevent is distinct and directly related to the main event. For instance, under the main event of a volleyball match outcome, a subevent could be "Čeští volejbalisté vyhráli na turnaji v portugalském Matosinhos v základní skupině nad Kubou 3:0."
- 3. Determine Timing:**
 - For **time_start** and **time_end** of the main event, use the date from the query if available. If no specific date is mentioned, use the year if it can be inferred from the context. If no date information is present, use "NA".
 - For subevents, if they are directly tied to the main event's timing, use "PARENT". If a subevent has its own specific timing, use the appropriate date format (YYYY-MM-DD). For example:

```
{
  "event": "Čeští volejbalisté vyhráli na turnaji v portugalském Matosinhos v základní skupině nad Kubou 3:0.",
  "subevents": [],
  "time_start": "NA",
  "time_end": "NA",
  "time_reported": "2018-06-24"
}
```

- 4. Set Reported Time:** The **time_reported** is taken directly from the query's date.

a) How would you select the optimal prompt? Is this an instance of Empirical Risk Minimization (ERM) learning?

b) How large should the benchmark set (i.e., how many examples (m)) be in order to guarantee, with 90% confidence, that the selected prompt's performance on unseen texts is at most 5% worse (in the proportion of good summaries) than the performance of the best prompt in the candidate set?

Solution 2. a) You have a set of P candidate prompts. Each prompt is evaluated on a benchmark consisting of m examples, and its empirical accuracy – the fraction of good summaries –

is computed. The prompt achieving the highest empirical accuracy is then selected. This procedure is an instance of the Empirical Risk Minimization (ERM) learning algorithm applied to a finite hypothesis class with P hypotheses (prompts) to choose from.

b) The required sample size to achieve a Probably Approximately Correct (PAC) guarantee with parameters $\varepsilon = 0.05$ and $\delta = 0.1$ is given by

$$m_{\text{PAC}}^{\mathcal{H}}(\varepsilon, \delta) = \frac{2}{\varepsilon^2} \log \left(\frac{2P}{\delta} \right) = \frac{2}{0.05^2} \log \left(\frac{2 \times 100}{0.1} \right) \approx 6,081 .$$

Hence, the benchmark must contain at least 6,081 examples to ensure, with 90% confidence, that the selected prompt's performance on unseen data is within 5 of the best possible prompt.